# Adaptive Privacy Budget Allocation in Federated Learning: A Multi-Agent Reinforcement Learning Approach

Zejian Chen, Guocheng Liao, *Member, IEEE,* Qian Ma, *Member, IEEE,* and Xu Chen, *Senior Member, IEEE*

***Abstract*—Federated learning is a popular distributed machine learning paradigm that keeps data locally at clients. To further enhance privacy protection, differential privacy techniques are incorporated in the federated learning framework. We can quantify the privacy budget (or privacy protection level) through differential privacy and allocate the budget to different communication rounds according to the composition property of differential privacy. Recent works have shown that suitably allocating budgets to different iterations can improve model performance. How to allocate privacy budgets in different communication rounds for different clients in the federated learning framework is a significant problem to study. The problem is challenging to solve due to the unknown relationship between noise levels and the model accuracy and the coupling property of the clients' decisions. In this paper, we propose a method based on multi-agent reinforcement learning to solve the privacy budget allocation problem, which maximizes the accuracy of the federated learning model given limited privacy budgets for the clients. The experiments show that our proposed method is better than the uniform allocation, arithmetic sequence allocation, and exponential allocation methods.**

***Index Terms*—federated learning, deep reinforcement learning, differential privacy**

## I. Introduction

Nowadays, many machine learning aided applications (e.g., object detection [1], medical image analysis [2], smart building [3], and games [4]) facilitate people's lives. The centralized training of machine learning models requires clients to send their data to a central server, which imposes the risk of data privacy leakage. Instead, federated learning [5], proposed by Google, is a distributed learning framework. It allows the clients to train a shared machine learning model by only uploading the local model updates without sharing their data.

However, privacy leakage still exists in federated learning. Existing studies (e.g., [6], [7]) have pointed out that the uploaded local model still contains information regarding private training data, leading to potential privacy leakage. To further enhance privacy protection, some studies (e.g., [8], [9]) have proposed differentially-private federated learning frameworks, in which the clients can add random noises into their local models to prevent their training data from being distinguished by the adversaries.

In differentially-private federated learning, clients obfuscate their local models by a certain noise level according to their *privacy budgets*. We will later quantify the privacy budget by a differential privacy parameter. According to the composition property [10] of differential privacy, the client can flexibly allocate his budget to different communication rounds of federated learning, as long as the sum of budgets of all rounds does not exceed his predefined total privacy budget. Existing studies (e.g., [11]) suggested that allocation methods significantly impact the final model's performance. Specifically, the gradient is large at the early stages of the training and, thus, can tolerate a high noise level. However, as the training proceeds, the gradient becomes smaller, and a high noise level could significantly degrade the model. This observation indicates an opportunity to carefully allocate the privacy budget for each training round to optimize the model performance, given the total privacy budget constraint.

Many existing works have studied privacy budget allocation methods. Fan *et al.* in [12] and Xie *et al.* in [13] proposed arithmetic sequence allocation methods for the k-means algorithm and deep learning model, respectively, and achieved an effective balance between training accuracy and data privacy. Zhang *et al.* in [14] considered adopting an exponential allocation method on deep learning models and achieved considerable performance on medical datasets. Lee *et al.* in [15] studied the privacy budget allocation method based on the utility of privacy-preserving statistics and showed that the method works well with multiple versions of differential privacy. These studies show the possibility of achieving better model performance by allocating different privacy budgets at different iterations. Motivated by this, we study the privacy budget allocation problem in federated learning.

The privacy budget allocation problem in federated learning has the following challenges. Firstly, the exact relationship between noise levels and model accuracy is unclear. Some existing studies (e.g., [8], [16]–[18]) have analyzed the performance guarantee in terms of convergence upper bound of differentially-private federated learning. However, to our knowledge, the analytic relationship between noise levels and the exact model performance is still unknown. Secondly, federated learning is a multi-agent scenario. The privacy budget

Zejian Chen and Guocheng Liao are with the School of Software Engineering, Sun Yat-sen University, Zhuhai, China (Email: chenzj46@mail2.sysu.edu.cn; liaogch6@mail.sysu.edu.cn). Qian Ma is with School of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen, China (Email: maqian25@mail.sysu.edu.cn). Xu Chen is with the School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou, China (Email: chenxu35@mail.sysu.edu.cn).

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allocation decisions of different clients are coupled with each other. That is, the accuracy of the global model is affected by all clients' budget allocation decisions. In this case, one individual's contribution to the global model's accuracy will be diluted, making it challenging to adjust clients' decisions.

Considering the above challenges, we propose to use a method based on multi-agent reinforcement learning [19]. Reinforcement learning is an interactive machine learning method. Deep reinforcement learning (DRL) agents gain experience by interacting with the environment and use the experience to enhance their policies to take more intelligent actions in different states. Due to the exploratory nature of reinforcement learning, DRL agents can learn the relationship between noise intensity and deep learning model accuracy changes in continuous interactions, thereby making excellent privacy budget allocation decisions.

Our contributions are summarized as follows:

- We study a privacy budget allocation problem in a differentially-private federated learning framework. This problem is important because suitable allocation schemes can help improve the model performance while satisfying a total privacy budget constraint. The problem is challenging to solve due to the unknown relationship between model accuracy and the privacy budget and the coupling allocation decisions of clients in federated learning.
- To solve the above problem, we propose a multi-agent reinforcement learning-based algorithm to make privacy budget allocation decisions. To speed up the DRL training process, we adopt parameter sharing and value decomposition network techniques. We also use an action mapping module that adjusts the privacy allocation decisions to fully utilize the privacy budget.
- We perform extensive experiments and compare our method to other privacy budget allocation methods, including the uniform allocation, the arithmetic sequence allocation, and the exponential allocation. Our method can make better use of the privacy budget and outperforms the baselines, with an accuracy improvement of 1% on the MNIST dataset.

The remainder of this paper is organized as follows. We present the system model in Section II. We propose our method in Section III and show experimental results in Section IV. Finally, we make a summary in Section V.

## II. System Overview

We present the system model of differentially-private federated learning in detail in this section. We first briefly introduce the traditional federated learning framework in Section II-A. Then, we present the threat model and how to use differential privacy techniques to protect clients' privacy in Section II-B. Finally, we formulate a privacy budget allocation problem in Section II-C.

### A. Federated Learning

Federated learning is a decentralized learning framework in which the clients train a shared model by only uploading the local model updates to the server. This allows clients to conduct machine learning without revealing their private datasets. We consider a set $\mathcal{K} = \{1, 2, ..., K\}$ of $K$ clients. Each client $k \in \mathcal{K}$ has its private dataset $\mathcal{D}_k = \left\{\left(\mathbf{x}_k^1, y_k^1\right), \left(\mathbf{x}_k^2, y_k^2\right), ..., \left(\mathbf{x}_k^{|\mathcal{D}_k|}, y_k^{|\mathcal{D}_k|}\right)\right\}$, where $\mathbf{x}_k^i$ and $y_k^i$ represent the feature vector and the label, respectively, of $i$th sample in his dataset $\mathcal{D}_k$.

The process of federated learning is as follows. At first, the server initializes a global model with parameters $\theta^0$. For each communication round $t$, $t = 0, 1, ..., T-1$, all the clients receive the global model $\theta^t$ from the server. Then each client $k \in \mathcal{K}$ performs mini-batch gradient descent using his $J$ disjoint mini batches $\left\{\mathcal{B}_k^1, ..., \mathcal{B}_k^J | \bigcup \mathcal{B}_k^j = \mathcal{D}_k, \mathcal{B}_k^i \bigcap \mathcal{B}_k^j = \emptyset, \forall 1 \leq i, j \leq J\right\}$:

$$\theta_k^t \leftarrow \theta_k^t - \frac{\eta}{\left|\mathcal{B}_k^i\right|} \sum_{\left(\mathbf{x}_k^j, y_k^j\right) \in \mathcal{B}_k^i} \nabla \mathcal{L}\left(\mathbf{x}_k^j, y_k^j; \theta_k^t\right), \forall i = 1, ..., J, \tag{1}$$

for $E$ epochs to find model parameters $\theta_k^t$ that minimize the local loss function $\mathcal{L}\left(\mathcal{D}_k; \theta_k^t\right)$ on his private dataset. Then the server aggregates the local models $\theta_k^t$ from all clients $k \in \mathcal{K}$ and obtains the global model $\theta^{t+1}$:

$$\theta^{t+1} \leftarrow \frac{\sum_{k=1}^{K} |\mathcal{D}_k| \, \theta_k^t}{\sum_{k=1}^{K} |\mathcal{D}_k|}. \tag{2}$$

The loop is terminated when the current communication round $t$ reaches a predefined limit $T$.

### B. Differential privacy

We assume the server is honest-but-curious [20], which is a common privacy threat assumption. In this assumption, the legitimate server will not deviate from the federated learning protocol but will attempt to learn possible information from legitimately uploaded model parameters.

To reduce the risk of privacy leakage, the clients in the federated learning framework can choose to inject noise into their local models. At the same time, we can use differential privacy techniques to analyze the privacy protection level for the clients. Here, we use one variant of differential privacy framework, $\rho$-zero-concentrated differential privacy ($\rho$-zCDP) [10] for privacy accounting due to its simple combination theorem and its relatively tight privacy budget bound.

**Definition 1 ($\rho$-zCDP [10])** *For any two adjacent datasets $D, D' \in \mathcal{D}$ which differ in only one single entry, A randomized mechanism $\mathcal{M}: \mathcal{D} \to \mathbb{R}$ is $\rho$-zCDP if for all $\alpha \in (1, \infty)$,*

$$R_\alpha\left(\mathcal{M}(D) || \mathcal{M}\left(D'\right)\right) \leq \rho\alpha, \tag{3}$$

*where $R_\alpha(P||Q)$ is the Rényi diversity which is defined as*

$$R_\alpha(P||Q) = \frac{1}{\alpha - 1} log \mathbb{E}_{x \sim Q}\left(\frac{P(x)}{Q(x)}\right)^\alpha, \tag{4}$$

*given random variables $P$ and $Q$.*

We use the parameter $\rho$ to quantify the *privacy budget*. When $\rho$ is lower, we can know from the inequality (3) that the



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**Algorithm 1:** Differentially-private Federated Learning

**Input :** client datasets $\{\mathcal{D}_k\}$, global model $\theta^0$, clipping bound $C$, epochs for local updating $E$, noise level $\{\sigma_k^t\}$, total communication rounds $T$, and learning rate $\eta$

1 **for** $t = 0, 1, ..., T-1$ **do**
2  Each client receives the last global model $\theta^t$ from the server;
3  **for** $k \in \mathcal{K}$ **do**
4  $\theta_k^t \leftarrow \theta^t$;
5  **for** $e \in \{1, ..., E\}$ **do**
6  Client $k$ divides his dataset $\mathcal{D}_k$ into mini-batches $\left\{\mathcal{B}_k^j\right\}$;
7  **for** $j \in \{1, 2, ..., J\}$ **do**
8  **for** *each sample* $(\mathbf{x}_k, y_k) \in \mathcal{B}_k^j$ **do**
9  Calculate the gradient $\nabla\mathcal{L}\left(\mathbf{x}_k, y_k; \theta_k^t\right)$;
10  Clip the gradient $\bar{\nabla}\mathcal{L}\left(\mathbf{x}_k, y_k; \theta_k^t\right) \leftarrow \nabla\mathcal{L}\left(\mathbf{x}_k, y_k; \theta_k^t\right) \times \min\left(1, \frac{C}{||\nabla\mathcal{L}(\mathbf{x}_k, y_k; \theta_k^t)||_2}\right)$;
11  **end**
12  Add Gaussian noise to the mean of gradients $\widetilde{\nabla}\mathcal{L}\left(\mathcal{B}_k^j; \theta_k^t\right) \leftarrow \frac{1}{B_k}\left(\sum \bar{\nabla}\mathcal{L}\left(\mathbf{x}_k, y_k; \theta_k^t\right) + \mathcal{N}\left(0, \left(\sigma_k^t\right)^2 C^2 \mathbf{I}\right)\right)$;
13  Gradient descent $\theta_k^t \leftarrow \theta_k^t - \eta\widetilde{\nabla}\mathcal{L}\left(\mathcal{B}_k^j; \theta_k^t\right)$;
14  **end**
15  **end**
16  **end**
17  Server collects and aggregates local models from clients $\theta^{t+1} \leftarrow \frac{\sum_{k=1}^K |\mathcal{D}_k| \theta_k^t}{\sum_{k=1}^K |\mathcal{D}_k|}$;
18 **end**

**Output:** $\theta^T$

output probabilities on the adjacent datasets are more similar, which makes it more difficult for an adversary to infer a client's private dataset. A higher value of $\rho$ indicates weaker privacy protection or more serious privacy leakage. Thus, we can leverage parameter $\rho$ to characterize a client's privacy budget that he is willing to contribute to federated learning.

To achieve differential privacy, we consider adding Gaussian random noise to the gradients [8] before gradient descent. We describe the differentially-private federated learning in Algorithm 1. Unlike traditional federated learning, in Lines 9-10, each client calculates the gradient and performs clipping for each sample in the mini-batch. In Line 12, each client adds Gaussian noise as per noise level $\{\sigma_k^t\}$ to the clipped gradients to provide privacy protection.

**Theorem 1** *For each client $k \in \mathcal{K}$, with batch size $B_k$ and noise levels $\{\sigma_k^t\}$, Algorithm 1 guarantees $\left(\sum_{t=0}^{T-1} \frac{2E}{(\sigma_k^t)^2}\right)-zCDP$.*

Theorem 1 shows that when noise levels $\{\sigma_k^t\}$ are higher, the privacy protection is stronger. Combining the fact that higher noise levels can cause larger deviations in the gradients of federated learning models, we can see a trade-off between privacy protection and learning performance. Besides, we also notice that there are different choices of noise levels $\{\sigma_k^t\}$ (giving different learning performance) that satisfy the same total privacy budget. This suggests we can flexibly allocate the privacy budget to each communication round subject to the total budget constraint to increase the learning performance, which will be discussed in the next section.

### C. Privacy Budget Allocation Problem

According to Theorem 1, different privacy budget allocations can achieve the same privacy protection. This motivates us to adjust the privacy budget allocation of each communication round to improve model accuracy. In this regard, we formulate a privacy budget allocation problem for each client $k$ to optimize the performance of the federated learning model.

$$\begin{aligned} \max_{\{\sigma_k^t\}} \quad & Acc\left(\bigcup_{k=1}^K \mathcal{D}_k; \theta^T\right) \\ s.t. \quad & \sigma_{min} \le \sigma_k^t \le \sigma_{max}, t = 0, ..., T-1, \\ & \sum_{t=0}^{T-1} \rho_k^t \le \rho_k, \\ & \rho_k^t = \frac{2E}{(\sigma_k^t)^2}, t = 0, ..., T-1, \end{aligned} \tag{5}$$

where $[\sigma_{min}, \sigma_{max}]$ is the interval of the selected noise levels and $\rho_k$ is the total privacy budget of client $k$.

Solving the problem in (5) is challenging. Firstly, the relationship between decision variables and the optimization objective is difficult to obtain. Secondly, the accuracy of the global model is affected by all clients' decisions. Such coupling property further complicates the problem.

We propose a multi-agent reinforcement learning-based method to solve this problem. During exploratory interaction, DRL agents can implicitly learn the relationship between noise level and model accuracy. In addition, the credit assignment technique in multi-agent reinforcement learning, which we will explain in the next section, can help decouple the decision of each client.

## III. The Proposed Method

In this section, we introduce our method based on multi-agent reinforcement learning to solve the problem in (5). We first describe the decentralized partially observable Markov decision process formulation in Section III-A. We then introduce the design of our multi-agent reinforcement learning algorithm in detail in Section III-B.



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### A. Decentralized Partially Observable Markov Decision Process

We first transform the scenario into a decentralized partially observable Markov decision process (Dec-POMDP) [21] $\mathcal{G} = \left(\mathcal{K}, \{\mathcal{S}_k\}_{k\in\mathcal{K}}, \{\mathcal{A}_k\}_{k\in\mathcal{K}}, R, P, \gamma\right)$. Here, $\mathcal{K}$ is the set of $K$ clients, $\mathcal{S}_k$ represents the state locally observable by client $k$, $\{\mathcal{A}_k\}_{k\in\mathcal{K}}$ are the sets of action spaces, $R : \mathcal{S}_1 \times ... \times \mathcal{S}_K \times \mathcal{A}_1 \times ... \times \mathcal{A}_K \to \mathbb{R}$ is the reward function of the clients, $P : \mathcal{S}_1 \times ... \times \mathcal{S}_K \times \mathcal{A}_1 \times ... \times \mathcal{A}_K \times \mathcal{S}_1 \times ... \times \mathcal{S}_K \to [0, 1]$ is the transition probability, and $\gamma$ represents the discount factor. Specifically, we define state, action, and reward as follows:

**State**: $s_k^t = (t, \rho_k)$. Here, $t$ is the current communication round and $\rho_k$ is the total privacy budget of client $k$.

**Action**: $a_k^t \in [\sigma_{min}, \sigma_{max}]$ is a 1-dimensional action that determines the noise level $\sigma_k^t = a_k^t$ for the current communication round $t$.

**Reward**: The reward $r_t = Acc\left(\bigcup_{k=1}^{K} \mathcal{D}_k; \theta^{t+1}\right) - Acc\left(\bigcup_{k=1}^{K} \mathcal{D}_k; \theta^{t}\right)$ is defined as the accuracy difference between adjacent communication rounds.

### B. Multi-agent Reinforcement Learning Algorithm Design

*1) Deep Deterministic Policy Gradient:* We use the deep deterministic policy gradient (DDPG) algorithm [22] to train the clients' policies. The DDPG algorithm is a deep reinforcement learning algorithm based on the actor-critic framework and can effectively handle continuous action spaces. Each client has two networks, i.e., the critic network $Q(s, a)$ and the policy network $\pi(s)$. The critic network $Q(s, a)$ evaluates how good the action $a$ is given the state $s$, and the policy network $\pi(s)$ represents the current policy that selects an action given the state $s$.

*2) Parameter Sharing:* To reduce the computational cost and speed up the DRL training process, we introduce the parameter sharing technique [23] to our network architecture, i.e., all clients share the same parameters in their policy networks and critic networks. In this way, we only need to train one policy network and one critic network.

*3) Value Decomposition Network:* The reward is designed as a single global reward, which cannot precisely reflect the contribution of different clients and hinders the process of training DDPG. To evaluate the contributions of different clients, we add a value-decomposition network [24], which sums up the outputs of all clients' critic networks.

*4) Action Mapping:* We design an action mapping module to adjust privacy budget allocation decisions to fully utilize clients' privacy budgets. According to Theorem 1, if $\sum_{t=0}^{T-1} \rho_k^t < \rho_k$, part of total budget $\rho_k$ is not utilized, which could lead to model performance degradation. To fully utilize the privacy budget (i.e., $\sum_{t=0}^{T-1} \rho_k^t = \rho_k$ exactly holds), we propose an affine mapping to map an underutilized budget allocation $\{\rho_k^t\}$ to a fully-utilized budget allocation $\{a_k\rho_k^t + b_k\}$. Here, $a_k$ and $b_k$ are the parameters of the affine mapping. Meanwhile, such mapping also helps accelerate the training of DDPG by eliminating unnecessary exploration of underutilization allocation. To derive the suitable affine mapping, we formulate a squared error minimization problem subject to full budget utilization for each client $k \in \mathcal{K}$.

$$\begin{aligned} \min_{a_k, b_k} \quad & \sum_{t=0}^{T-1} \left(\rho_k^t - a_k\rho_k^t - b_k\right)^2 \\ s.t. \quad & \frac{2E}{\sigma_{max}^2} \le a_k\rho_k^t + b_k \le \frac{2E}{\sigma_{min}^2}, \forall t = 0, ..., T-1, \\ & \sum_{t=0}^{T-1} \left(a_k\rho_k^t + b_k\right) = \rho_k. \end{aligned} \tag{6}$$

**Algorithm 2:** DDPG Algorithm

**Initialize:** Critic network $Q_\phi(s, a)$ with parameters $\phi$ and policy network $\pi_\psi(s)$ with parameters $\psi$, target networks $Q_{\phi'}$ and $\pi_{\psi'}$ with $\phi' \leftarrow \phi, \psi' \leftarrow \psi$, replay buffer $\mathcal{R} \leftarrow \emptyset$, and soft update rate $\tau$

**for** $episode = 1, ..., M$ **do**
  **for** $k \in \mathcal{K}$ **do**
    Select actions $\{a_k^t\}$ with $a_k^t = \pi_\psi(s_k^t) + \mathcal{N}$, where $\mathcal{N}$ is a Gaussian noise;
    Perform action mapping and recalculate new actions $\{\bar{a}_k^t\}$;
  **end**
  **for** $t = 0, ..., T-1$ **do**
    **for** $k \in \mathcal{K}$ **do**
      Execute action $\bar{a}_k^t$ in Algorithm 1;
    **end**
    Store transitions $\left\{\left(s_k^t, \bar{a}_k^t, r^t, s_k^{t+1}\right)\right\}_{k\in\mathcal{K}}$ in $\mathcal{R}$;
    Sample minibatch of transitions $\left\{\left(s_k^i, \bar{a}_k^i, r^i, s_k^{i+1}\right)\right\}$ with size $L$ from $\mathcal{R}$;
    Calculate

$$y^i = r^i + \gamma \sum_{k\in\mathcal{K}} Q_{\phi'}\left(s_k^{i+1}, \pi_{\psi'}\left(s_k^{i+1}\right)\right);$$

    Update the critic network with value decomposition network by minimizing the loss function:

$$\mathcal{L} = \frac{1}{L}\sum_i \left(y^i - \sum_{k\in\mathcal{K}} Q_\phi\left(s_k^i, \bar{a}_k^i\right)\right)^2;$$

    Update the policy network by policy gradient:

$$\nabla J_\psi \approx \frac{1}{K}\frac{1}{L}\sum_k\sum_i \nabla_a Q_\phi\left(s_k^i, a\right)|_{a=\pi_\psi(s_k^i)} \times \nabla_\psi \pi_\psi\left(s_k^i\right);$$

    Update target networks:

$$\phi' \leftarrow \tau\phi + (1-\tau)\phi', \psi' \leftarrow \tau\psi + (1-\tau)\psi';$$

  **end**
**end**



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| | MNIST | CIFAR-10 |
|---|---|---|
| Ours | **91.05%** | **47.75%** |
| Uniform | 88.03% | 46.49% |
| Exponential | 90.02% | 47.59% |
| Arithmetic sequence | 90.04% | 47.43% |

TABLE I: Model accuracy under different privacy budget allocation schemes on MNIST and CIFAR-10 datasets.

**Proposition 1** *Problem (6) is a convex optimization problem. The solutions $a_k^*$ and $b_k^*$ can be obtained by KKT conditions.*

*5) Algorithm:* The whole DDPG algorithm is shown in Algorithm 2. Firstly, in Lines 2-5, the clients select actions based on the current policy network and correct the actions with the action mapping module. In Lines 7-10, the clients interact with the environment, i.e., Algorithm 1 to collect transitions. The algorithm updates the critic network with the TD error formulation in Lines 11-13 and updates the policy network with policy gradient in Line 14. Finally, the algorithm updates the target networks with the soft update formulas in Line 15.

## IV. Evaluation

### A. Experimental Setup

We consider $N = 10$ clients. We run our experiments on two datasets: MNIST and CIFAR-10. We use different settings for two datasets. 1) MNIST setting: each client's total privacy budget is randomly sampled from $[2, 4]$. The batch size is $B_k = 200$, the number of epochs for local updating is $E = 4$, the clipping bound is $C = 4$, the range of noise levels $[\sigma_{min}, \sigma_{max}]$ is $[1, 20]$, the total communication rounds is $T = 20$ and the local learning rate is $\eta = 0.1$. 2) CIFAR-10 setting: each client's total privacy budget is randomly sampled from $[20, 40]$. The batch size is $B_k = 200$, the number of epochs for local updating is $E = 4$, the clipping bound is $C = 6$, the range of noise levels $[\sigma_{min}, \sigma_{max}]$ is $[1, 20]$, the total communication rounds is $T = 25$, and the local learning rate is $\eta = 0.1$. The federated learning model for both settings has two convolutional layers and three fully connected layers. We test 100 times on the test sets for both settings.

For the multi-agent reinforcement learning algorithm, both the policy network and the critic network are composed of three fully connected layers. We use a sigmoid activation function to constrain the outputs of the policy network and map these outputs into the action space. The episodes for the MNIST setting and the CIFAR-10 setting are 200 and 350, respectively.

For comparison, we choose the uniform allocation method [8], the arithmetic sequence allocation method [13], and the exponential allocation method [14] as the baselines of our experiments.

### B. Experimental Results

Fig. 1 shows the model accuracy achieved by four allocation methods on two datasets. We can see that our method outperforms the other three baselines. Although our method is inferior to the other three baselines at the early stage of training (e.g., communication round $t \leq 10$), it manages to catch up with other baselines at the final stage (e.g., communication round $t \geq 19$). This shows that our method can flexibly allocate the privacy budget to increase the model performance. Meanwhile, Table I shows the final accuracy of four allocation methods. We can see that when compared with the baselines, our method can increase the accuracy by around 1% and 0.2% on the MNIST dataset and CIFAR-10 dataset, respectively.

Figure. 2 shows the noise levels of different communication rounds for clients of different total privacy budgets. We can see from both Fig. 2a and Fig. 2b that the noise level decreases as the training proceeds. This is because the magnitude of the gradient at the early stage of training is larger than that at the latter stage. This indicates that the training can tolerate a higher noise level at the earlier stage. Our method has learned to save privacy budget in the early stage and use it aggressively in the later stage. Furthermore, by comparing different curves in each subfigure of Fig. 2, we can see that when a client's total privacy budget is higher, the noise level he imposes in each communication round is lower. This shows that our method can adaptively allocate different privacy budgets based on the clients' privacy demands.

## V. Conclusion

There is potential to improve the performance in the differentially-private federated learning framework by allocating privacy budgets carefully. We study the privacy budget allocation problem in the differentially-private federated learning scenario, which is a difficult problem because of decision coupling and the unclear relationship between the decisions and the optimization goal. To solve the problem, we propose a method based on multi-agent reinforcement learning with parameter sharing, value decomposition network, and action mapping. By using the DDPG algorithm to allocate the privacy budget to each communication round, the accuracy of federated learning can be improved. The experiments show that the proposed method outperforms some other baselines.

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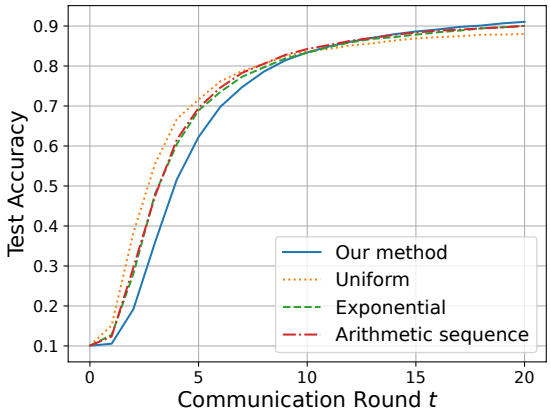
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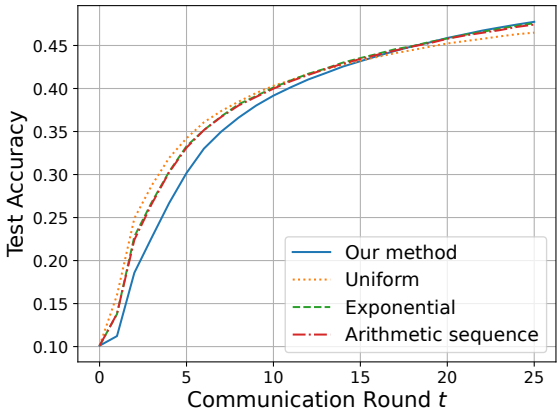
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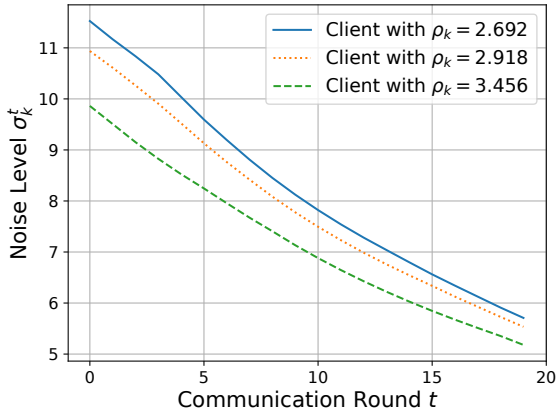



(a) Accuracy on MNIST dataset.

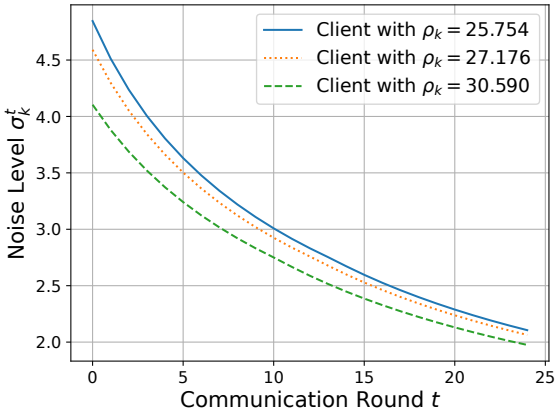



(b) Accuracy on CIFAR-10 dataset.

Fig. 1: Comparison of model accuracy under different privacy budget allocation schemes on MNIST and CIFAR-10 datasets.




(a) Noise levels of three clients with different total privacy budgets on MNIST dataset.




(b) Noise levels of three clients with different total privacy budget on CIFAR-10 dataset.

Fig. 2: Noise levels of different communication rounds on MNIST and CIFAR-10 dataset.


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